Team Tam Slam - Need4Data F26
Project Synopsis

Throughout the duration of the week, our team , Team Tam Slam, has cleaned, analyzed, and visualized 500,000 recorded U.S. traffic accidents from 2016 to 2023 to investigate how time of day, weather conditions, geographic location, and road infrastructure are associated with accident severity and traffic disruptions. We placed ourselves into the shoes of an Emergency Services Department Analytics Team to propose how we can make the roads safer to reduce crash severity. In order to do so, we used Python and it's libraries such as Pandas, Matplotlib, Numpy, and Seaborn to clean and integrate data from four analytical categories (Time, Weather, Geographic Location, and Traffic Infrastructure), reducing the original 46 variables to 31 selected and derived variables while preserving all 500,000 observations. We conducted exploratory data analysis and developed visualizations to identify patterns in accident frequency and severity. Our findings showed higher recorded accident counts during morning and afternoon commuting hours, geographic concentrations of reported accidents, and differences in high-severity accident proportions across weather conditions. For example, approximately 22.6% of accidents recorded during rain were classified as Severity 3 or 4, compared with approximately 17.2% during clear weather. To continue with, we also examined associations between roadway infrastructure features and recorded severity. Based on all these findings, we proposed targeted road-safety strategies involving traffic management, weather alerts, and infrastructure evaluation, while acknowledging limitations related to missing observations, unequal category sizes, and the observational nature of the dataset!